



XMBA 200S: Data and Decisions

Chapter 15: Confidence Intervals

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Data and Decisions Part 1:

The data you want versus the data you've got

Managers (and others) almost never have all of the data they want...

- On their customers
- On their production facilities
- On their employees' productivity

If you can only collect **a sample of data**

- What can you say if you have **no information at all** about a process?
- We are building toward the next section of the course: **statistical testing** – decision-making with no information about the population or any process

The set up for today

- Last class we discussed the distribution of sample mean
 - There is a distribution of sample means centered at population mean
 - Sample mean has a standard deviation (standard error) equal to the population s.d. divided by $\sqrt{n}$
- We also discussed how often we would see very unusual sample means (calculate a z-score)
- In this class, we use this info to pin down where the population mean is likely to be using data from a sample

Outline

- Confidence intervals for population parameters
- Margin of error and sample size determination
- Practice with data: in-class Excel exercise

UC Berkeley Alumni Credit Card



- Before offering a UC Berkeley alumni credit card, a credit card company performs a pilot study.
- Preapproved applications are sent to a random sample of 1000 UC Berkeley alumni.
- On the basis of the results of the pilot the company decides whether or not to launch the card.

Parameters of Interest

What Do We Need to Know?

- The proportion who will return the application for the credit card. Let's call this p .
- The average monthly balance that those who accept the credit card will carry. Let's call this μ .

Summary statistics

Number of offers	1,000
Number accepted	140
Proportion who accepted	$\hat{p} = 0.14$
Average (balance)	$\bar{x} = \$1,990.50$
SD (balance)	$s = \$2,833.33$
Kurtosis K_4	2.975

Confidence Intervals

Introduction

- A confidence interval is a range of plausible values for a parameter based on one sample.
- Constructing confidence intervals relies on the sampling distribution of the statistic.
- To construct a confidence interval you need to know:
 - How confident you want to be (α)
 - The mean in your sample ($\bar{x}$)
 - The standard error (S.E.)

Reminder: what is the “standard error”

- The **standard error (SE) of a sample average** tells us how far off the sample average tends to be from the truth
- We calculate it using the following formula (or software 😊)

$$SE = \frac{\sigma}{\sqrt{n}}$$

- σ is the population standard deviation
- n is the sample size

The more typical case: What if we don't know the standard deviation from the population?

Implementation

- In most cases, σ is not observed...
- Instead, the analyst (or software) substitutes the sample standard deviation (s) in for the population standard deviation (σ) to construct the standard error.

$$SE = \frac{s}{\sqrt{n}}$$

Confidence interval: formula

A confidence interval is a range of plausible values for a parameter based on one sample.

Formula:

The $100(1 - \alpha)\%$ confidence interval for μ is defined as:

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \times S.E.$$

or, writing out the definition of the standard error:

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \times \frac{s}{\sqrt{n}}$$

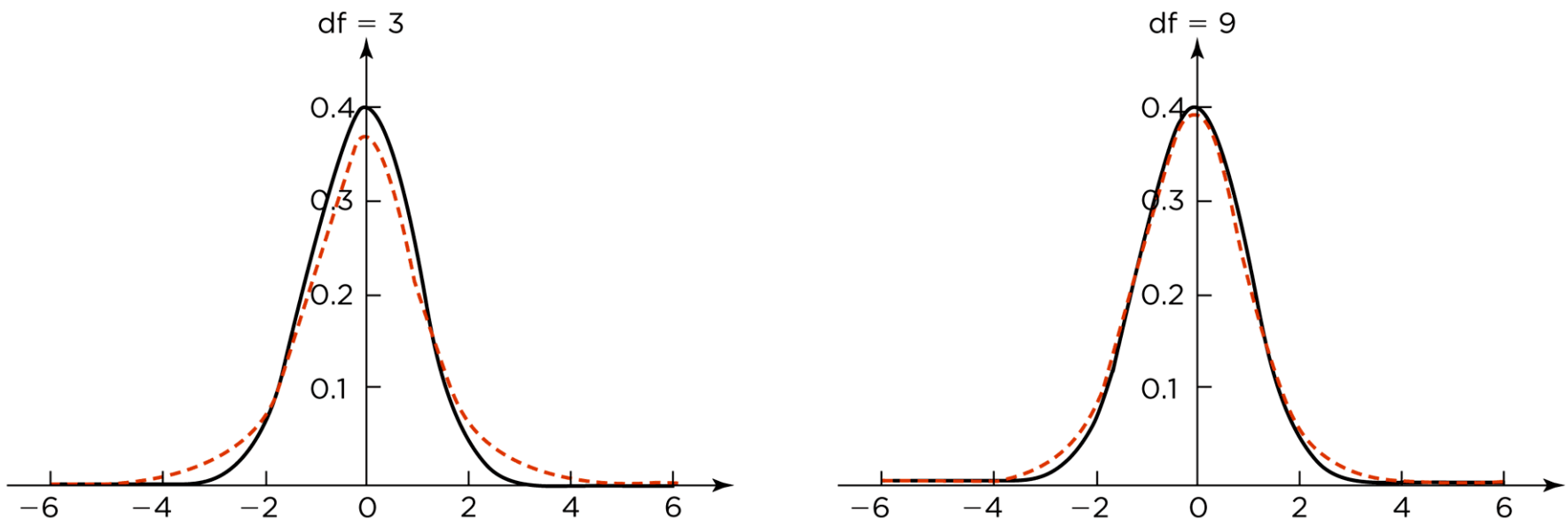
Confidence Interval for the Mean

Student's t-Distribution

- A “normal looking” distribution but with wider tails.
- Used when we substitute sample standard deviation s for population standard deviation σ in the standard error.
- The *degrees of freedom* ($df = n-1$) controls the shape of the distribution.
- As n increases, the t-distribution more closely resembles the standard normal distribution.

Confidence Interval for the Proportion

Student's t-Distribution



Standard normal (black) and Student's t-distributions (red)

Calculating T-Statistics

You chose a 95% confidence level, as implied by $\alpha = 0.05$

You have $n = 35$

You want to find the t-statistic from a distribution with 34 degrees of freedom that puts .025 in each tail $t_{\frac{\alpha}{2}, n-1} = t_{0.025, 34}$

Excel = T.INV(*probability*, df) = T.INV(0.025, 34) = 2.032

Stata: “display invttail(34, 0.025)” = 2.032

Confidence Interval for the Proportion

Credit Card Example: Proportion of Offers Accepted

The estimated standard error is,

$$\frac{s}{\sqrt{n}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.14 (1 - 0.14)}{1000}} = 0.01097$$

This formula for the standard error makes sense because we already saw that the standard error of the sample mean is $\sigma/\sqrt{n}$ and the population variance for a proportion is $p(1 - p)$. For more on proportions see Chapter 11.

Thus the 95% confidence interval is,

$$0.14 \pm 1.96 (0.01097) = [0.1185 \text{ to } 0.1615]$$

We are 95% confident that p lies between 11.8% and 16.2%.

Ranges for Parameters

Credit Card Example

- With 95% confidence, the population proportion that will accept the offer is between about 12% and 16%.
 - Can we do better?
- A larger sample size will reduce the estimated standard error resulting in a narrower interval (a more precise estimate of p).

Confidence Interval for the Mean

Introduction

- A similar procedure as that used for p is used to construct a confidence interval for μ .
- The estimated standard error for the sampling distribution of the sample mean is $\frac{s}{\sqrt{n}}$.
- Here we substitute s for σ , just like we substituted $\hat{p}$ for p .

Confidence Interval for the Mean

Credit Card Example

The estimated standard error is:

$$\frac{s}{\sqrt{n}} = \frac{\$2,833.33}{\sqrt{140}} = \$239.46$$

Thus the 95% confidence interval is

$$\begin{aligned} & \$1,990.50 \pm 1.98 (\$239.46) \\ & [\$1,516 \text{ to } \$2,464] \end{aligned}$$

We are 95% confident that μ lies between \$1,516 and \$2,464.

Confidence Interval for the Mean

Credit Card Example

We are 95% confident that μ lies between \$1,516 and \$2,465.

Might μ be \$1,250? It could be, but based on the sample results it's not likely.

Interpreting confidence intervals

Incorrect interpretations

- 95% of all customers keep a balance of \$1,520 to \$2,460
- The mean balance of 95% of samples of 140 accounts will fall between \$1,520 and \$2,460
- The mean balance is between \$1,520 and \$2,460

How Does AI Do Here?

Number of offers	1,000
Number accepted	140
Proportion who accepted	$\hat{p} = 0.14$
Average (balance)	$\bar{x} = \$1,990.50$
SD (balance)	$s = \$2,833.33$
Kurtosis K_4	2.975

calculate the confidence interval for the proportion and then for the mean

Thought for 6s >

CI for the proportion (all 1,000 offers):

$$SE(\hat{p}) = \sqrt{(0.14 \times 0.86 / 1000)} = \sqrt{0.0001204} \approx 0.01097$$

$$95\% \text{ CI: } 0.14 \pm 1.96 \times 0.01097 = 0.14 \pm 0.0215 \rightarrow \mathbf{(0.1185, 0.1615)}$$

So between about 11.9% and 16.2% of customers accept the offer. (Conditions check out: $n\hat{p} = 140$ and $n(1-\hat{p}) = 860$, both well above 10.)

CI for the mean balance (the 140 who accepted):

Note $n = 140$ here, not 1,000 — the balance data only exists for accepters.

$$SE(\bar{x}) = 2833.33 / \sqrt{140} = 2833.33 / 11.832 \approx \$239.46$$

$$\text{Using } t \text{ with } 139 \text{ df } (t^* \approx 1.977): 1990.50 \pm 1.977 \times 239.46 = 1990.50 \pm 473.42 \rightarrow \mathbf{(\$1,517, \$2,464)}$$

With $z = 1.96$ you'd get $(\$1,521, \$2,460)$ — essentially identical at this sample size.

Outline

- Confidence intervals for population parameters
- Margin of error and sample size determination
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Margin of Error

Definition

The margin of error is an informal, back-of-the-envelope 95% confidence interval that replaces the statistic from the z or t distribution with 2.

- For the sample mean, the margin of error is $2\frac{s}{\sqrt{n}}$
- For the proportion, the margin of error is $2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

This is commonly used with election polls and other surveys. For example, Gallup might report that 46% approve of the president with a margin of error of ± 2 percentage points.

Margin of Error

- A precise confidence interval has a small margin of error.
- Margin of error is affected by:
 1. level of confidence
 2. variation in the data
 3. number of sample observations.
- It is used in determining sample size.

Determining sample size

- Margin of Error is $2 \times se(\bar{X})$, where 2 is an approx. to 1.96
- For a study about μ with 95% coverage, find the sample size using

$$n = 4\sigma^2 / (\text{Margin of Error})^2$$

- Obtain an estimate for σ^2 using a pilot sample (since we have to choose n before collecting data)

Determining sample size when studying proportions

- For a study about p , no need for a pilot sample
- Use $p = 0.5$ which results in largest possible value for σ of $\frac{1}{2}$
- So, the necessary sample size is just:

$$n = 1 / (\text{Margin of Error})^2$$

Margin of Error

Example.

- A nutritionist wants to know the average calorie intake for female customers to within ± 50 calories with 95% confidence. A pilot study gives an estimate of 430 calories for σ . Find n .

$$n = 4(430^2) / 50^2 = 295.8 \text{ or } 300 \text{ customers}$$

Determining sample size when studying proportions

Sample sizes for various margins of error
(95% coverage)

<i>n</i>	Margin of Error
100	10%
400	5%
625	4%
1,112	3%
2,500	2%
10,000	1%

Caution: Margin of Error w/ Non-Response

How the Polls Got it So Wrong in Wisconsin

Francesca Hong, the democratic socialist running in Wisconsin's Democratic primary for governor, had a double-digit lead in polls as recently as last week. Then she lost. What happened?

See Chapter 14
slides on
Samples with
Selection,
Mitigation #5
(Bounds) + Link
on bCourses

Applications and Case Studies

Using Total Margin of Error to Account for Non-Sampling Error in Election Polls

Jeff Dominitz & Charles F. Manski

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Abstract

The potential impact of non-sampling errors on election polls is well known, but measurement has focused on the *margin of sampling error*. Statisticians have recommended measurement of total survey error by mean square error (MSE), which jointly measures sampling and non-sampling errors. We suggest use of the square root of maximum MSE to measure the *total margin of error* (TME). We suggest that measurement of TME should be a standard feature in the reporting of polls. Because the exceedingly low response rates commonly obtained by polls is a particularly worrisome source of potential error, we demonstrate how to measure the potential impact of nonresponse by sampled persons using the concept of TME. We show how to measure TME when a pollster lacks any knowledge of the candidate preferences of nonrespondents. We extend the analysis to settings where the pollster has partial knowledge that bounds the preferences of non-respondents. In each setting, we derive a simple poll estimate that approximately minimizes TME—a *midpoint estimate*—and compare it to a conventional poll estimate. We extend our analysis of nonresponse by sampled persons to address polls with under-coverage of the population and opt-in panels of participants.

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Caution: Margin of Error w/ Non-Response

- Aug 13, 2026: “How the Polls Got It So Wrong in Wisconsin” (NYT). A double-digit lead, then a loss. The polls said ± 3 .
- Prompt Fable: “1,250 voters contacted, 1,000 responded, 60% support candidate A. Write the confident client summary with the margin of error.”
 - Fable reports $60\% \pm 3$ — and never mentions the 250 who didn’t answer.
 - ± 3 assumes nonrespondents look exactly like respondents
- Worst Case Bounds: $0.8 \times 60\% + 0.2 \times [0-100\%] = [48\%, 68\%]$ — width of ignorance = nonresponse rate
- Total margin of error $\approx \pm 10$ (midpoint estimate) — sampling error is the trivial part; n fixes the small problem
- Real political polls: $\sim 2.5\%$ response. Now what do the bounds say?

In-Class Drill: The Too-Narrow Interval

Aug 13, 2026: “How the Polls Got It So Wrong in Wisconsin” (NYT). A double-digit lead, then a loss. The polls said ± 3 .

Prompt Fable: “1,250 voters contacted, 1,000 responded, 60% support candidate A. Write the confident client summary with the margin of error.”

Fable reports $60\% \pm 3$ — and never mentions the 250 who didn’t answer.

- ± 3 assumes nonrespondents look exactly like respondents
- No-assumptions bounds: $0.8 \times 60\% + 0.2 \times [0-100\%] = [48\%, 68\%]$ — width of ignorance = nonresponse rate
- Total margin of error $\approx \pm 10$ (midpoint estimate) — sampling error is the trivial part; n fixes the small problem
- Real political polls: $\sim 2.5\%$ response. Now what do the bounds say?

Example: Confidence Intervals in Action

Philip Morris vs. U.S. EPA

- The Environmental Protection Agency released a report in 1993 called, “Respiratory Health Effects of Passive Smoking”.
- The study concluded that secondhand smoke causes lung cancer in nonsmokers, and designated it a “Group A carcinogen” along with e.g. asbestos, radon, and benzene, and thus subject to increased regulation.
- A key piece of evidence in the study was an estimate that exposure to second hand smoke decreases lifespans by 1.6 years relative to average. This was based on a study of thousands of non-smokers who live with smokers.
- The EPA reported that the 90% confidence interval for this estimate is 0.12 to 3.08 years.

Example: Confidence Intervals in Action



Medicare Fraud

- CMS/Medicare audits claims from providers to prevent overpayment
- Audits are random samples from provider claims (stratified, etc...) n=100 out of 10,000+ claims
- The sample's overpayment rate (a statistic!) is extrapolated to the whole universe of provider claims
 - Program Integrity Manual, Ch. 8
- The provider is “fined” the lower end of the 90% confidence interval on that overpayment rate (applied to all claims!)
 - Palm Valley Health Care v. Azar (5th Cir. 2020)
 - \$81,681 found in a 54-claim sample → \$12 million demanded

There are lawyers and statisticians whose entire job is arguing about whether a confidence interval was built correctly — with eight-figure stakes

Example: Confidence Intervals in Action

Philip Morris vs. U.S. EPA

- Philip Morris and RJR Nabisco sued the EPA, charging that the findings were “manipulated” to “falsely disparage” cigarettes, asking for the report to be declared “null and void”.
- The battle centered on the “90% confidence interval”. What do you think the tobacco interests claimed?



PHILIP MORRIS



Example: Confidence Intervals in Action

- The tobacco interests argued that epidemiological studies always use a 95% confidence interval, and that in *choosing* a 90% interval instead they falsely represented the precision of their results.
- Indeed, if the EPA had instead used a 95% confidence interval, the range would have been wider.
 - The ratio of $t_{.025, n=large}$ to $t_{.05, n=large}$ is, $1.96 / 1.645 = 1.19$.
 - Thus the 95% confidence interval is about 20% wider than the 90% confidence interval.
 - The 95% confidence interval is -0.16 to 3.36 years.
- Why do you suppose that the EPA reported a 90% confidence interval instead? How do you feel about this choice? Did the EPA “manipulate” the results?

Example: Confidence Intervals in Action

Philip Morris vs. U.S. EPA

- Many scientists responded in support for the EPA. “The scientific case against environmental tobacco smoke is now overwhelming,” claimed an editorial in the Journal of National Cancer Institute.
- Moreover, the EPA argued that they didn’t want to use a 95% confidence interval because it couldn’t possibly be true that passive smoke decreases cancer risk, and that any hint of this would have made the report confusing.

The United States District Court ruled in favor of the tobacco industry in 1998.

- Ruled that the EPA had failed to follow proper scientific and epidemiologic practices and had "cherry picked" evidence to support their conclusions.
- *“The EPA publicly committed to a conclusion before research had begun...and adjusted established procedure and scientific norms to validate the Agency's public conclusion”.*

Punchlines

- You now have a way to determine *precisely*:
 1. When does a sample indicate an outcome unusual enough that you should act
 2. What does a sample indicate about a parameter that you don't know
- What's next: Making **decisions** based only on samples: statistical testing

Outline

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